

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-012 - Digboi-012

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000133
Field / District	Digboi / Tinsukia
Block	PEL-ASSAM-01
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2019-12-14 / 2020-03-19
Measured / TVD	3052 m / 3031 m
Trajectory	Vertical; inc 2.1 deg; az 41.2 deg
Formation Interval	1571-2298 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	3236 psi
Fracture Gradient	15.60 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	High (58.1/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	763	Surface
9 5/8 in casing	1770	Intermediate
7 in liner/casing	2685	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2020-03-10 2799 m	Mud Loss	Critical	Flow-out dropped below flow-in with pit loss. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2020-01-18 1246 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Escalate connection-gas trends before entering correlated pressure zone.
2020-01-26 1498 m	Torque Spike	High	High torque developed while WOB remained elevated. Mitigation: Reduce WOB/RPM and circulate before resuming normal parameters. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2020-02-25 2379 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Condition hole and confirm pressure normalization. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2020-02-25 2398 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2020-02-27 2453 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2020-01-12 1070 m	Mud Loss	Medium	Active pit volume declined during stable pump rate. Mitigation: Optimize hydraulics and maintain conservative ROP through loss interval. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.

Completion Summary

The synthetic well reached 3052 m MD in the Barail interval. The NWIS knowledge record links 14 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.